1-) Project Name

Analyzing Global CO2 Emissions and the Renewable Energy Transition for SDG7 and SDG13

2-) Overview

This project aims to analyze the complex, interconnected system of global energy consumption, economic development, and environmental impact. By utilizing the "Global Data on Sustainable Energy" dataset, we will decode the multivariate relationships driving carbon emissions across different nations. The core of this analysis is fundamentally aligned with two critical United Nations Sustainable Development Goals:

- **SDG 7 (Affordable and Clean Energy):** This goal emphasizes ensuring access to affordable, reliable, sustainable, and modern energy, highlighting the urgent need to increase the share of renewable energy in the global energy mix.
- **SDG 13 (Climate Action):** This goal demands urgent action to combat climate change and its impacts, primarily focusing on the critical reduction of global greenhouse gas (CO2) emissions.

Problem Statement: For decades, rapid economic expansion (GDP growth) and increasing electricity access in developing nations have been rigidly coupled with fossil fuel consumption. This structural dependency has created a severe bottleneck, resulting in exponential growth in global carbon emissions. The transition to clean energy is not a uniform process; it is heavily constrained by macroeconomic realities, infrastructure limitations, and inadequate financial flows. Traditional analyses often view environmental metrics in isolation, failing to address the multi-variable trade-offs between a nation's necessary economic development and strict global emission limits. Consequently, there is a critical need for a mathematically sound, system level framework that understands how structural variables collectively influence a nation's capacity to adopt renewable energy sources.

Structural Evaluation: To address this complex problem, the project moves beyond basic reporting through rigorous data preprocessing and structural evaluation. We aim to identify critical bottlenecks in the transition to clean energy and uncover systemic patterns in emission trajectories between 2000 and 2020. For instance, the analysis will critically investigate whether increased access to electricity in developing economies strictly correlates with higher fossil fuel reliance, or if targeted financial support effectively accelerates their shift to sustainable alternatives.

Impact Potential: Decoding these systemic interactions is not just an analytical exercise; it transforms raw global energy data into a robust, structurally optimized foundation essential for advanced AI applications and Operations Research models. By establishing a clean, scaled, and highly correlated system matrix, this evaluation sets the precise groundwork required for future multi-objective optimization algorithms. Ultimately, these future models will be capable of identifying optimal energy policies and resource allocation strategies. This will allow policymakers to maximize renewable energy integration under strict environmental constraints, directly contributing to actionable progress towards SDG 7 and SDG 13.

3-) Background

Achieving sustainable development on a global scale requires navigating the intricate trade-offs between economic expansion and environmental preservation. For decades, industrial growth and increasing access to electricity have been intrinsically linked to the consumption of fossil fuels, resulting in exponential growth in global greenhouse gas emissions. Today, the urgent need to mitigate climate change has catalyzed a worldwide transition towards renewable energy sources. However, this transition

is highly complex and non-uniform; it is heavily influenced by a country's macroeconomic conditions, technological infrastructure, and the availability of international financial flows for clean energy.

Traditional approaches to measuring environmental impact often look at emission metrics in isolation. However, to truly understand the multi dimensional nature of global energy dynamics, a more comprehensive system analysis perspective is required. The "Global Data on Sustainable Energy" dataset provides a critical twenty-year window (2000-2020) to observe these complex interactions. By analyzing this extensive historical data, we can move beyond basic reporting to uncover the underlying structural patterns of how developing and developed nations manage their energy portfolios. Understanding these structural dependencies is a crucial first step. It allows us to map the intricate network of macroeconomic and environmental variables, laying the necessary groundwork for deploying advanced artificial intelligence models designed to optimize resource allocation and emission reduction strategies under strict environmental constraints.

4) Key Objectives / Business Objectives

a) Research Questions:

1. How can macroeconomic indicators (e.g., GDP per capita, population density) and clean energy access be mathematically mapped to evaluate structural bottlenecks in global carbon emissions?
2. How do international financial flows structurally impact the transition to renewable energy sources (SDG 7), specifically within developing economies?
3. What is the optimal objective function required to balance rapid economic growth with strict climate action constraints and global emission limits (SDG 13)?
4. How does the historical trajectory (2000-2020) of energy integration demonstrate the systemic trade-offs between fossil fuel reliance and sustainable development?
5. What are the systemic efficiency gains of directing targeted financial interventions toward specific infrastructure bottlenecks, compared to untargeted energy investments?

b) Key Steps:

1. Design and implement a robust data preprocessing pipeline to structure and map macroeconomic indicators into a unified energy transition framework.
2. Develop and implement structural evaluation mechanisms to identify bottlenecks in national renewable energy capacities.
3. Analyze multivariate relationships and structural correlations to evaluate the trade-offs between economic growth and environmental preservation at various development stages.
4. Test the validity of the structural dependencies using historical data across various geographic and economic regions.
5. Evaluate the performance of different national energy portfolios against global sustainability benchmarks (SDG 7 and SDG 13).
6. Prepare findings and explore real world policy applications.

5) Methods and Workflow

a) Datasets:

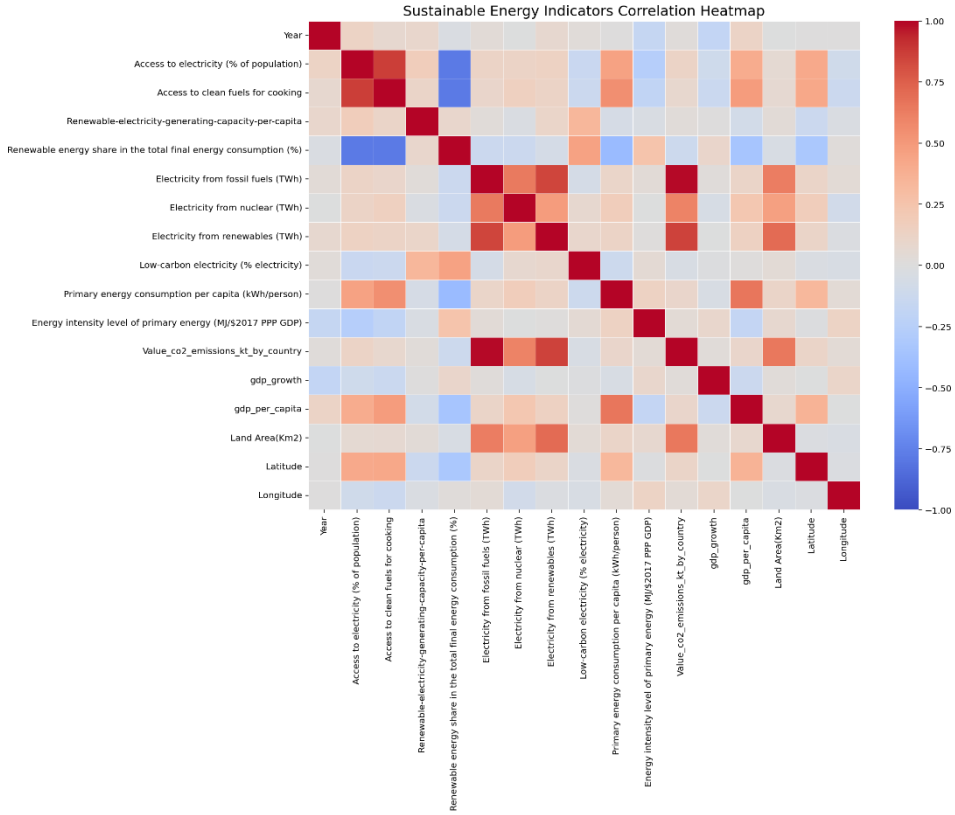
- **Source:** "Global Data on Sustainable Energy" encompassing macroeconomic, environmental, and infrastructure metrics.
- **Scope:** The dataset is utilized to map the systemic interactions between energy access, renewable integration, and CO2 emissions across different nations and years.

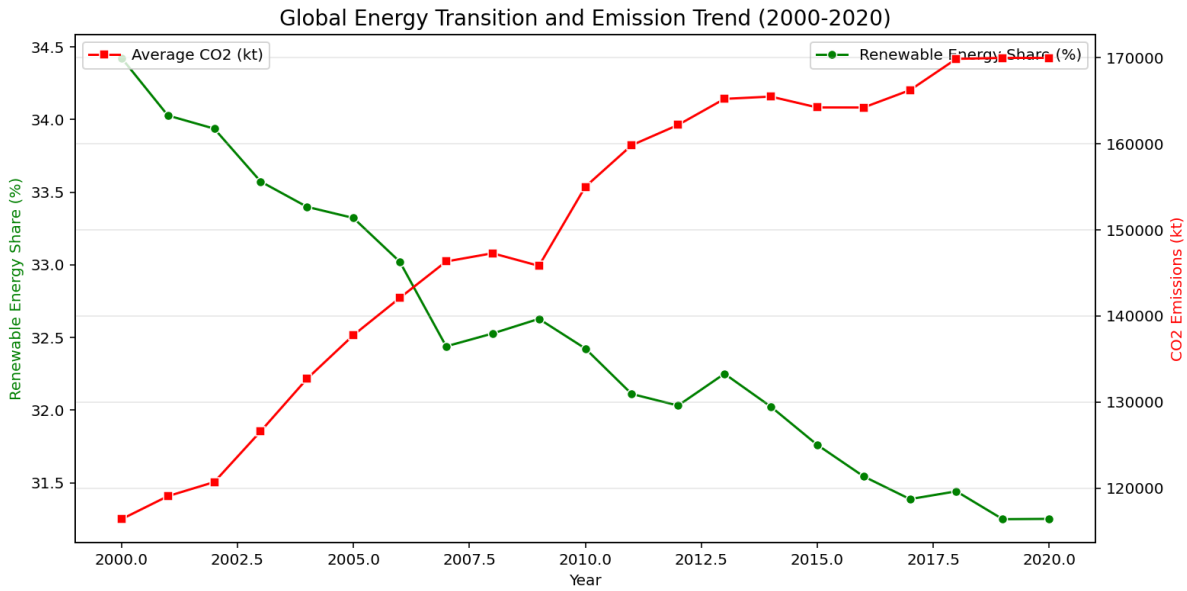
b) Data Cleaning and Preprocessing:

1. **Advanced Imputation:** Addressing missing values in critical numerical columns by applying bi-directional linear interpolation, followed by median fill techniques to ensure the structural integrity of the time series data without dropping essential rows.
2. **Data Transformation and Scaling:** Standardizing the systemic matrix to allow for balanced, multi-variable evaluation:
   - Applying Box-Cox transformation to the target variable (CO2 emissions) to stabilize variance.
   - Utilizing Min-Max Scaler for percentage-based infrastructure metrics (e.g., Access to electricity, Renewable energy share).
   - Utilizing Standard Scaler for absolute macroeconomic indicators (e.g., GDP per capita, Primary energy consumption) to ensure uniform mathematical weighting.

c) Modelling (EDA Phase):

1. **Multivariate Correlation Analysis:** Generating a comprehensive correlation matrix (heatmap) to decode the relationships and dependencies between economic growth and environmental indicators.
2. **Automated Feature Selection (Multicollinearity Control):** Implementing a dynamic threshold (0.85) to identify and eliminate highly correlated variables within the upper triangle of the correlation matrix, thereby reducing system noise while strictly protecting the target emission variable.
3. **Distribution and Trend Analysis:** Computing descriptive statistics and skewness to analyze variable distributions.
4. **Temporal Visualization:** Mapping the historical trajectory (2000-2020) of the global energy transition by plotting the integration of renewable energy against the macroscopic CO2 emission trends.





d) Deliverables:

1. A functional Python codebase demonstrating the rigorous data preprocessing, imputation, and feature selection pipeline.
2. Visual reports including structural correlation heatmaps and temporal trend analyses.
3. A fully cleaned, transformed, and scaled dataset, structurally optimized for advanced operations research and multi-objective optimization algorithms.

Emre Başoğlu

Barkın Atmaca